

EECS-841

Image Processing and Computer Vision

Final Examination, Session: Spring 25

Student's name:

ID:

Date:

Exam Guidelines:

1. You have 150 minutes to complete this exam.
2. The exam is closed book. No cheat-sheet is allowed.
3. Don't keep any book, cell phone, tab, laptop, copies, notebook at your desk while you are attending the exam.
4. Detail work/explanation on problems is suggested to receive partial credit.
5. Complete all your answers within the given answer sheet. No additional page will be provided.
6. The exam is meant to test your understanding. So be patient and read the questions/problems carefully before you answer.
7. If you don't submit your copy on time/when asked, you will get a penalty of 5 pts reduction for every additional minute delay.
8. It is not advised to leave the classroom during the exam.

DO NOT do anything that might be perceived as cheating, such as:

- Sharing solutions with each other.
- Getting help from anyone other than yourself.
- If any type of cheating is detected, the minimum penalty is getting zero for the exam. There are more academic consequences because this misconduct will be reported to the department chair.

Scoring:

Question No.	Points allocated	Points obtained by student
1	10	
2	15	
3	15	
4	15	
5	15	
6	10	
7	10	
8	10	
Total = 100 points		

Total points obtained by student:

Assumptions for all the questions:

- If I is an image, $I(x,y)$ denotes the pixel intensity at (x,y) location where x denotes column index and y denotes row index.
- The index starts from 0 for both the column and row.

Question 1. Min filters are commonly used in the pooling layer of CNNs. Apply a Min filter (with a size of 2×2) on image A with a stride of 2 and show the results. (10 points)

2	4	3	7
1	12	13	5
0	3	5	2
7	3	11	7

Image A

Answer:

Question 2. Perform correlation on the input image utilizing the given 3x3 kernel and show the output image, ensuring that the output image also maintains the same size as the input image. (15 points)

1	1	1	0
0	1	1	1
1	0	1	1
0	0	1	1

Input image

1	0	1
1	0	1
1	0	1

kernel

Answer:

Question 3. A convolutional neural network (CNN) is a popular architecture for classification tasks in computer vision. Illustrate a basic CNN architecture, highlighting the major components (ensure they are labelled in your diagram). (5 points)

Alice wants to train a CNN for dog vs cat classes; her dataset has grayscale images of size 25 x 25. Bob is also trying to build a CNN for dog vs. cat classification; however, the dataset he is using has images of size 100 x 100. Who do you think might be more likely to use a bigger stride and kernel size? Justify your answer for both stride and kernel size choices. (5+5=10 points)

Answer:

Question 4. Autoencoders and Vision Transformers are also promising deep learning models for classification tasks. Briefly describe the key idea behind their architectures. (5+5 = 10 points)

You are designing a classification pipeline for a resource-constrained device on a medical imaging dataset where each image must be classified as either benign or malignant. Which architecture between Autoencoder and Transformer do you think will be a better choice and why? (5 points)

Answer:

Question 5. What is the full form of “HoG”-based feature descriptor? Write down the step-by-step HoG-based feature extraction process. *(10 points)*

In the context of object recognition, how may the performance of a HoG-based feature extractor differ when applied to textured versus non-textured objects? Can you give an example of which type of image is likely to yield more informative features, and why? *(5 points)*

Answer:

Question 6. What is the key idea behind the watershed-based segmentation method? Can you state the major differences between marker-based and non-marker-based methods for segmentation? (5+ 5 = 10 points)

Answer:

Question 7. SVM is one of the commonly used traditional machine learning methods for classification. Below are the classification criteria for a traditional SVM model for binary classification:

The decision function for a linear SVM is:

$$f(x) = w^T x + b$$

- x = input feature vector (e.g., image features)
- w = weight vector (learned by SVM)
- b = bias term

classification rule is:

If $f(x) \geq 0$, predict class $+1$

If $f(x) < 0$, predict class -1

Suppose you want to build an SVM model for classifying three different classes (e.g., car, pedestrian, and tree), for object detection purposes in autonomous vehicles. Now, consider the base model of the SVM classifier again (given above). Do you think it is possible to classify among the three classes utilizing the model as it is? Why or why not? If your answer is not, then please also share a strategy on how to build a new SVM model or modify/utilize the given model for performing the three-class classification. (10 points)

Answer:

Extra Page